

# Aditya Sathishkumar

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## EDUCATION

### University of Massachusetts Amherst

Amherst, MA

Bachelor of Science in Computer Science

Expected May 2027

Coursework – 3.84 GPA

Data Structures, Algorithms, Statistics, Object-Oriented Programming, Information Visualization, Linear Algebra

### Extracurricular

Research (**CAD modeled** cooling paper and generated heat maps to analyze radiance and energy coverage), Machine Learning Club (Trained self-balancing agent with **reinforcement learning** and **control algorithms**)

## EXPERIENCE

### Software Engineer Intern

Aug 2025 – Current

NASA

Merritt Island, FL

- Engineered **Python automated test suite** validating **9,000+ flight-critical requirements**, boosting coverage and reducing manual verification time
- Developing end-to-end **systems verification** framework linking requirements to implementations with **MySQL** and **REST API** for traceability
- Built **data ingestion pipeline** converting **XML scripts** into structured logs, accelerating **fault analysis**
- Transformed **test outputs** into analyzable datasets on Linux, enabling rocket **anomaly detection** and reliability

### Software Engineer Co-op

Jan 2025 – Jul 2025

Huntington Ingalls Industries (HII)

Dayton, OH

- Developed real-time **C++** training pipeline for in-flight **Machine Learning**, reducing development time **80%**
- Optimized gimbal control for automated aircraft tracking with **Boost** and **GUI** libraries to process **5 Gbps**
- Engineered RF receiver using **Software-defined Radio**, **C**, and **Multithreading**, reducing latency **35%**
- Programmed **C++ API** for Chrono timekeeping to improve precision **6 decimals** and reduce data overlap **20%**
- Simulated flight tracks in **C++ Engine** for custom playback and Machine Learning output validation
- Executed unit, simulation, memory leak, and field tests to ensure system reliability and error recovery
- Collaborated in cross-functional team using **Git**, **Agile**, **Linux** and **Visual Studio**

### Software Engineer

Sep 2023 – Dec 2023

UMass Robotics

Amherst, MA

- Designed **C++ Computer Vision** navigation for ROS-based rover, reducing depth error **20%**
- Leveraged **Asynchronous** messaging for **Computer Vision** and **Inertial** sensor fusion and **2x** fault tolerance
- Employed image processing with **OpenCV** to reduce system execution time and resource consumption **25%**

## PROJECTS

### My JobScraper | Python, BeautifulSoup, Selenium

Jun 2024

- Programmed **Python Script** to Web Scrape career pages and compiled **2,000+** available positions to automate students' job search process, allowing company and role searching
- Personalized job search results by employing Selenium and BeautifulSoup for page navigation, pagination, parsing, and pattern recognition, reducing application time by **50%**

### University Weather | JavaScript, Node.js, Git, Jest, RESTful APIs

Nov 2024

- Developed app backend to fetch weather and coordinates of 1,000+ colleges by querying **APIs** using JS and **Git** to inform users of weather at their respective schools
- Tested comprehensively using Jest framework for **test-driven development**, ensuring integration and accuracy

## SKILLS

**Languages/Dependencies:** C++, C, Python, ImGui, Boost, OpenCV, Requests, GTest, Jest

**Technologies:** Git, Gitlab, Visual Studio, Object-Oriented Design, Computer Vision, Version Control, Data Pipelines, System Verification, Communication, Collaboration, Linux, Windows, Machine Learning, Testing, Integration, Simulation, API, Networking, Agile